

$$B = \lim_{x \rightarrow +\infty} \left(\frac{x^2 + 1}{x} - \ln x \right) = \lim_{x \rightarrow +\infty} \left(x + \frac{1}{x} - \ln x \right) = \lim_{x \rightarrow +\infty} \left[x \left(1 + \frac{1}{x^2} - \frac{\ln x}{x} \right) \right]$$

ដោយ $\lim_{x \rightarrow +\infty} (x) = +\infty$ និង $\lim_{x \rightarrow +\infty} \left(1 + \frac{1}{x^2} - \frac{\ln x}{x} \right) = 1$ ព្រោះ $\lim_{x \rightarrow +\infty} \frac{\ln x}{x} = 0$ ។

$$\text{ដូចនេះ } B = \lim_{x \rightarrow +\infty} \left(\frac{x^2 + 1}{x} - \ln x \right) = +\infty \text{ ។}$$

$$C = \lim_{x \rightarrow +\infty} \left(-x + 1 - \frac{\ln x}{x} \right) = \lim_{x \rightarrow +\infty} \left[x \left(-1 + \frac{1}{x} - \frac{\ln x}{x^2} \right) \right]$$

$$= \lim_{x \rightarrow +\infty} (x) \lim_{x \rightarrow +\infty} \left(-1 + \frac{1}{x} - \frac{\ln x}{x^2} \right)$$

ដោយ $\lim_{x \rightarrow +\infty} (x) = +\infty$ និង $\lim_{x \rightarrow +\infty} \left(-1 + \frac{1}{x} - \frac{\ln x}{x^2} \right) = -1$ ព្រោះ $\lim_{x \rightarrow +\infty} \frac{\ln x}{x^2} = 0$ ។

$$\text{ដូចនេះ } \lim_{x \rightarrow +\infty} \left(-x + 1 - \frac{\ln x}{x} \right) = -\infty \text{ ។}$$

$$D = \lim_{x \rightarrow +\infty} \frac{x + \ln x}{2x - 3} \text{ មានរាងមិនកំណត់ } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow +\infty} \frac{x \left(1 + \frac{\ln x}{x} \right)}{x \left(2 - \frac{3}{x} \right)} = \lim_{x \rightarrow +\infty} \frac{1 + \frac{\ln x}{x}}{2 - \frac{3}{x}} = \frac{1}{2}$$

$$\text{ព្រោះ } \lim_{x \rightarrow +\infty} \frac{\ln x}{x} = 0 \text{ និង } \lim_{x \rightarrow +\infty} \frac{1}{x} = 0 \text{ ។}$$

$$\text{ដូចនេះ } D = \lim_{x \rightarrow +\infty} \frac{x + \ln x}{2x - 3} = \frac{1}{2} \text{ ។}$$

$$E = \lim_{x \rightarrow +\infty} \frac{2x - 1 + \ln x}{3x + 1 - \ln x} \text{ មានរាងមិនកំណត់ } \frac{\infty}{\infty}$$

$$= \lim_{x \rightarrow +\infty} \frac{x \left(2 - \frac{1}{x} + \frac{\ln x}{x} \right)}{x \left(3 + \frac{1}{x} - \frac{\ln x}{x} \right)} = \lim_{x \rightarrow +\infty} \frac{2 - \frac{1}{x} + \frac{\ln x}{x}}{3 + \frac{1}{x} - \frac{\ln x}{x}} = \frac{2}{3}$$

ព្រោះ $\lim_{x \rightarrow +\infty} \left(\frac{\ln x}{x} \right) = 0$ ដូចនេះ $E = \frac{2}{3}$ ។

$F = \lim_{x \rightarrow +\infty} \frac{2x^2 + x \ln x + 1}{(2x + 1)^2}$ មានរាងមិនកំណត់ $\frac{\infty}{\infty}$

$$= \lim_{x \rightarrow +\infty} \frac{x^2 \left(2 + \frac{\ln x}{x} + \frac{1}{x^2} \right)}{x^2 \left(2 + \frac{1}{x} \right)^2} = \lim_{x \rightarrow +\infty} \frac{2 + \frac{\ln x}{x} + \frac{1}{x^2}}{\left(2 + \frac{1}{x} \right)^2} = \frac{2}{4} = \frac{1}{2} \text{ ។}$$

ដូចនេះ $F = \lim_{x \rightarrow +\infty} \frac{2x^2 + x \ln x + 1}{(2x + 1)^2} = \frac{1}{2}$ ។

$G = \lim_{x \rightarrow +\infty} [2 \ln(2x + 1) - \ln(x^2 + 1)]$ មានរាងមិនកំណត់ $\infty - \infty$

$$= \lim_{x \rightarrow +\infty} [\ln(2x + 1)^2 - \ln(x^2 + 1)]$$

$$= \lim_{x \rightarrow +\infty} \ln \left[\frac{(2x + 1)^2}{x^2 + 1} \right] = \ln \left[\lim_{x \rightarrow +\infty} \frac{x^2 \left(2 + \frac{1}{x} \right)^2}{x^2 \left(1 + \frac{1}{x^2} \right)} \right]$$

$$= \ln \left[\lim_{x \rightarrow +\infty} \frac{\left(2 + \frac{1}{x} \right)^2}{1 + \frac{1}{x^2}} \right] = \ln 4 = 2 \ln 2 \text{ ។}$$

ដូចនេះ $G = \lim_{x \rightarrow +\infty} [2 \ln(2x + 1) - \ln(x^2 + 1)] = 2 \ln 2$ ។

$H = \lim_{x \rightarrow +\infty} [x + 3 - \ln(e^{x+1} + 2)]$ មានរាងមិនកំណត់ $\infty - \infty$

$$= \lim_{x \rightarrow +\infty} \left[x + 3 - \ln \left[e^x \left(e + \frac{2}{e^x} \right) \right] \right] = \lim_{x \rightarrow +\infty} \left[x + 3 - \ln(e^x) - \ln \left(e + \frac{2}{e^x} \right) \right]$$

$$= \lim_{x \rightarrow +\infty} \left[x + 3 - x - \ln \left(e + \frac{2}{e^x} \right) \right] = \lim_{x \rightarrow +\infty} \left[3 - \ln \left(e + \frac{2}{e^x} \right) \right] = 3 - \ln e = 2$$

ដូចនេះ $H = \lim_{x \rightarrow +\infty} [x + 3 - \ln(e^{x+1} + 2)] = 2$ ។

$I = \lim_{x \rightarrow +\infty} (\sqrt{\ln^2 x + 4 \ln x + 1} - \ln x)$ មានរាងមិនកំណត់ $\infty - \infty$

$$= \lim_{x \rightarrow +\infty} \frac{\ln^2 x + 4 \ln x + 1 - \ln^2 x}{\sqrt{\ln^2 x + 4 \ln x + 1} + \ln x} = \lim_{x \rightarrow +\infty} \frac{4 \ln x + 1}{\ln x \sqrt{1 + \frac{4}{\ln x} + \frac{1}{\ln^2 x}} + \ln x}$$

$$= \lim_{x \rightarrow +\infty} \frac{4 + \frac{1}{\ln x}}{\sqrt{1 + \frac{4}{\ln x} + \frac{1}{\ln^2 x}} + 1} = \frac{4}{1+1} = 2$$

ដូចនេះ $I = \lim_{x \rightarrow +\infty} (\sqrt{\ln^2 x + 4 \ln x + 1} - \ln x) = 2$ ។

$J = \lim_{x \rightarrow +\infty} [2x + 3 - \ln(e^{2x} - x)]$ មានរាងមិនកំណត់ $\infty - \infty$

ឧទាហរណ៍. គណនាលីមីតខាងក្រោម៖

$A = \lim_{x \rightarrow 0} \frac{\ln(1+x) - 2\ln(1-x)}{x}$

$B = \lim_{x \rightarrow 0} \frac{2\ln(1-2x) + 3\ln(1+4x)}{x}$

$C = \lim_{x \rightarrow 0} \frac{x \ln(1+2x) + 4 \ln(1+x^2)}{x^2}$

$D = \lim_{x \rightarrow 0} \frac{\ln(1+4x^2) - x \ln(1-2x)}{3x^2}$

$E = \lim_{x \rightarrow 0} \frac{\ln(1+2\sin x) - 2\ln(1-\sin 2x)}{x}$

$F = \lim_{x \rightarrow 0} \frac{\ln(2+x^2 - \cos 2x)}{x^2}$

$G = \lim_{x \rightarrow 0} \frac{\ln(2+x \sin 3x - \cos 2x)}{x^2}$

$H = \lim_{x \rightarrow 0} \frac{\ln(2e^{x^2} - \cos 4x)}{x^2}$

$I = \lim_{x \rightarrow 0} \frac{\ln(2 - e^{-x^2} \cos 2x)}{x^2}$

$J = \lim_{x \rightarrow 0} \frac{\ln(2 - \cos 2x \cos 4x)}{x^2}$

ដំណោះស្រាយ

គណនាលីមីតខាងក្រោម៖

$$A = \lim_{x \rightarrow 0} \frac{\ln(1+x) - 2\ln(1-x)}{x} \text{ មានរាងមិនកំណត់ } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} + 2 \lim_{x \rightarrow 0} \frac{\ln(1-x)}{-x} = 1 + 2 = 3 \text{ ។}$$

$$\text{ដូចនេះ: } A = \lim_{x \rightarrow 0} \frac{\ln(1+x) - 2\ln(1-x)}{x} = 3 \text{ ។}$$

$$B = \lim_{x \rightarrow 0} \frac{2\ln(1-2x) + 3\ln(1+4x)}{x} \text{ មានរាងមិនកំណត់ } \frac{0}{0}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\ln(1-2x)}{x} + 3 \lim_{x \rightarrow 0} \frac{\ln(1+4x)}{x}$$

$$= -4 \lim_{x \rightarrow 0} \frac{\ln(1-2x)}{-2x} + 12 \lim_{x \rightarrow 0} \frac{\ln(1+4x)}{4x} = -4 + 12 = 8$$

$$\text{ដូចនេះ: } B = \lim_{x \rightarrow 0} \frac{2\ln(1-2x) + 3\ln(1+4x)}{x} = 8 \text{ ។}$$

$$C = \lim_{x \rightarrow 0} \frac{x \ln(1+2x) + 4 \ln(1+x^2)}{x^2} \text{ មានរាងមិនកំណត់ } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{x \ln(1+2x)}{x^2} + \lim_{x \rightarrow 0} \frac{4 \ln(1+x^2)}{x^2}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\ln(1+2x)}{2x} + 4 \lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{x^2} = 2 + 4 = 6$$

$$\text{ដូចនេះ: } C = \lim_{x \rightarrow 0} \frac{x \ln(1+2x) + 4 \ln(1+x^2)}{x^2} = 6 \text{ ។}$$

$$D = \lim_{x \rightarrow 0} \frac{\ln(1+4x^2) - x \ln(1-2x)}{3x^2} \text{ មានរាងមិនកំណត់ } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1+4x^2)}{3x^2} - \lim_{x \rightarrow 0} \frac{x \ln(1-2x)}{3x^2}$$

$$= \frac{4}{3} \lim_{x \rightarrow 0} \frac{\ln(1+4x^2)}{4x^2} + \frac{2}{3} \lim_{x \rightarrow 0} \frac{\ln(1-2x)}{-2x} = \frac{4}{3} + \frac{2}{3} = 2$$

$$\text{ដូចនេះ: } D = \lim_{x \rightarrow 0} \frac{\ln(1+4x^2) - x \ln(1-2x)}{3x^2} = 2 \text{ ។}$$

$$E = \lim_{x \rightarrow 0} \frac{\ln(1+2\sin x) - 2\ln(1-\sin 2x)}{x} \text{ មានរាងមិនកំណត់ } \frac{0}{0}$$

$$= \lim_{x \rightarrow 0} \frac{\ln(1 + 2\sin x)}{x} - 2 \lim_{x \rightarrow 0} \frac{\ln(1 - \sin 2x)}{x}$$

$$= 2 \lim_{x \rightarrow 0} \frac{\ln(1 + 2\sin x)}{2\sin x} \times \frac{\sin x}{x} + 4 \lim_{x \rightarrow 0} \frac{\ln(1 - \sin 2x)}{-\sin 2x} \times \frac{\sin 2x}{2x} = 2 + 4 = 6$$

ដូចនេះ $E = \lim_{x \rightarrow 0} \frac{\ln(1 + 2\sin x) - 2\ln(1 - \sin 2x)}{x} = 6$ ។

$F = \lim_{x \rightarrow 0} \frac{\ln(2 + x^2 - \cos 2x)}{x^2}$ មានរាងមិនកំណត់ $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\ln(1 + x^2 + 2\sin^2 x)}{x^2 + 2\sin^2 x} \times \lim_{x \rightarrow 0} \frac{x^2 + 2\sin^2 x}{x^2}$$

$$= \lim_{x \rightarrow 0} \left(1 + \frac{2\sin^2 x}{x^2} \right) = 1 + 2 = 3$$

ដូចនេះ $F = \lim_{x \rightarrow 0} \frac{\ln(2 + x^2 - \cos 2x)}{x^2} = 3$ ។

$G = \lim_{x \rightarrow 0} \frac{\ln(2 + x \sin 3x - \cos 2x)}{x^2}$ មានរាងមិនកំណត់ $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\ln[1 + (x \sin 3x + 2\sin^2 x)]}{(x \sin 3x + 2\sin^2 x)} \times \lim_{x \rightarrow 0} \frac{x \sin 3x + 2\sin^2 x}{x^2}$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin 3x}{3x} \times 3 + 2 \times \frac{\sin^2 x}{x^2} \right) = 3 + 2 = 5$$

$H = \lim_{x \rightarrow 0} \frac{\ln(2e^{x^2} - \cos 4x)}{x^2}$ មានរាងមិនកំណត់ $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\ln(2e^{x^2} - 1 + 2\sin^2 2x)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\ln[1 + 2(e^{x^2} - 1 + \sin^2 2x)]}{2(e^{x^2} - 1 + \sin^2 2x)} \times \lim_{x \rightarrow 0} \frac{2(e^{x^2} - 1 + \sin^2 2x)}{x^2}$$

$$= 2 \lim_{x \rightarrow 0} \left[\frac{e^{x^2} - 1}{x^2} + \left(\frac{\sin 2x}{2x} \times 2 \right)^2 \right] = 2(1 + 4) = 10$$

ដូចនេះ $H = \lim_{x \rightarrow 0} \frac{\ln(2e^{x^2} - \cos 4x)}{x^2} = 10$ ។

$I = \lim_{x \rightarrow 0} \frac{\ln(2 - e^{-x^2} \cos 2x)}{x^2}$ មានរាងមិនកំណត់ $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\ln \left[1 + (1 - e^{-x^2} \cos 2x) \right]}{(1 - e^{-x^2} \cos 2x)} \times \lim_{x \rightarrow 0} \frac{1 - e^{-x^2} \cos 2x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1 - e^{-x^2} (1 - 2\sin^2 x)}{x^2} = \lim_{x \rightarrow 0} \frac{1 - e^{-x^2} + 2e^{-x^2} \sin^2 x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{e^{-x^2} - 1}{-x^2} + 2 \lim_{x \rightarrow 0} \left[e^{-x^2} \left(\frac{\sin x}{x} \right)^2 \right] = 1 + 2 = 3$$

ដូចនេះ $I = \lim_{x \rightarrow 0} \frac{\ln(2 - e^{-x^2} \cos 2x)}{x^2} = 3$ ។

$J = \lim_{x \rightarrow 0} \frac{\ln(2 - \cos 2x \cos 4x)}{x^2}$ មានរាងមិនកំណត់ $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\ln \left[1 + (1 - \cos 2x \cos 4x) \right]}{(1 - \cos 2x \cos 4x)} \times \lim_{x \rightarrow 0} \frac{1 - \cos 2x \cos 4x}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos 2x) + \cos 2x (1 - \cos 4x)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{2\sin^2 x + 2\cos 2x \sin^2 2x}{x^2}$$

$$= 2 \lim_{x \rightarrow 0} \left[\left(\frac{\sin x}{x} \right)^2 + \cos 2x \left(\frac{\sin 2x}{2x} \times 2 \right)^2 \right]$$

$$= 2(1 + 4) = 10$$

ដូចនេះ $J = \lim_{x \rightarrow 0} \frac{\ln(2 - \cos 2x \cos 4x)}{x^2} = 10$ ។

ឃ.លីមីតអាចមិនកំណត់ 1^∞

a) ព្រឹត្តិស្តីបទ

$$i) \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$$

$$ii) \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e \quad \text{ដែល } e = 2.71828... \text{ (គោលលោការីតនេព័រ)}។$$

b) ប្រែប្រួលទូទៅ

បើគេមានអនុគមន៍ពីរ f និង g ដែល $\lim_{x \rightarrow a} [f(x)] = 1$ និង $\lim_{x \rightarrow a} [g(x)] = \infty$

$$\text{នោះគេបាន } \lim_{x \rightarrow a} [f(x)]^{g(x)} = \left(e\right)^{\lim_{x \rightarrow a} \{g(x)[f(x)-1]\}} \quad \text{។}$$

ឧទាហរណ៍. គណនាលីមីតខាងក្រោម៖

$$A = \lim_{x \rightarrow 0} (1+4x)^{\frac{2}{x}}$$

$$B = \lim_{x \rightarrow 0} \left(\frac{1-x}{1+x}\right)^{\frac{1}{x}}$$

$$C = \lim_{x \rightarrow 0} (1+3\sin 4x)^{\frac{2}{x}}$$

$$D = \lim_{x \rightarrow 0} (2 - \cos 2x)^{\frac{1}{x \sin x}}$$

$$E = \lim_{x \rightarrow 0} (2e^x - 1)^{\frac{1}{x}}$$

$$F = \lim_{x \rightarrow 0} (2e^{x^2} - \cos 2x)^{\frac{1}{x^2}}$$

$$G = \lim_{x \rightarrow 0} \left(\frac{e^x + e^{3x}}{2}\right)^{\frac{1}{x}}$$

$$H = \lim_{x \rightarrow 0} \left(\frac{2\cos 2x}{1 + \cos 2x}\right)^{\frac{1}{x^2}}$$

$$I = \lim_{x \rightarrow 0} \left(\frac{1+x \sin 2x}{\cos 2x}\right)^{\frac{1}{x^2}}$$

$$J = \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x^2}}{\cos x}\right)^{\frac{1}{x^2}}$$

ដំណោះស្រាយ

គណនាលីមីតខាងក្រោម៖

$$A = \lim_{x \rightarrow 0} (1+4x)^{\frac{2}{x}} \quad \text{មានរាងមិនកំណត់ } 1^\infty$$

$$= \lim_{x \rightarrow 0} \left[(1 + 4x)^{\frac{1}{4x}} \right]^8 = e^8 \quad \text{។}$$

ដូចនេះ $A = \lim_{x \rightarrow 0} (1 + 4x)^{\frac{2}{x}} = e^8 \quad \text{។}$

$B = \lim_{x \rightarrow 0} \left(\frac{1-x}{1+x} \right)^{\frac{1}{x}}$ មានរាងមិនកំណត់ 1^∞

ដោយ $\frac{1-x}{1+x} = \frac{(1+x)-2x}{1+x} = 1 + \frac{-2x}{1+x}$ នោះគេបាន ៖

$$B = \lim_{x \rightarrow 0} \left\{ \left[1 + \left(\frac{-2x}{1+x} \right) \right]^{\frac{1+x}{-2x}} \right\}^{\frac{-2}{1+x}} = e^{-2} = \frac{1}{e^2} \quad \text{។}$$

ដូចនេះ $B = \lim_{x \rightarrow 0} \left(\frac{1-x}{1+x} \right)^{\frac{1}{x}} = \frac{1}{e^2} \quad \text{។}$

$C = \lim_{x \rightarrow 0} (1 + 3 \sin 4x)^{\frac{2}{x}}$ មានរាងមិនកំណត់ 1^∞

$$= \lim_{x \rightarrow 0} \left[(1 + 3 \sin 4x)^{\frac{1}{3 \sin 4x}} \right]^{\frac{6 \sin 4x}{x}} = (e)^{24 \lim_{x \rightarrow 0} \frac{\sin 4x}{4x}} = e^{24}$$

$D = \lim_{x \rightarrow 0} (2 - \cos 2x)^{\frac{1}{x \sin x}}$ មានរាងមិនកំណត់ 1^∞

$$= \lim_{x \rightarrow 0} (1 + 2 \sin^2 x)^{\frac{1}{x \sin x}} = \lim_{x \rightarrow 0} \left[(1 + 2 \sin^2 x)^{\frac{1}{2 \sin^2 x}} \right]^{\frac{2 \sin x}{x}} = e^2 \quad \text{។}$$

ដូចនេះ $D = \lim_{x \rightarrow 0} (2 - \cos 2x)^{\frac{1}{x \sin x}} = e^2$

$E = \lim_{x \rightarrow 0} (2e^x - 1)^{\frac{1}{x}}$ មានរាងមិនកំណត់ 1^∞

$$= \lim_{x \rightarrow 0} \left\{ \left[1 + 2(e^x - 1) \right]^{\frac{1}{2(e^x - 1)}} \right\}^{\frac{2(e^x - 1)}{x}} = e^{2 \lim_{x \rightarrow 0} \frac{e^x - 1}{x}} = e^2 \text{ ។}$$

ដូចនេះ $E = \lim_{x \rightarrow 0} (2e^x - 1)^{\frac{1}{x}} = e^2 \text{ ។}$

$F = \lim_{x \rightarrow 0} (2e^{x^2} - \cos 2x)^{\frac{1}{x^2}}$ មានរាងមិនកំណត់ 1^∞

$$= \lim_{x \rightarrow 0} (2e^{x^2} - 1 + 2\sin^2 x)^{\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0} \left\{ \left[1 + 2(e^{x^2} - 1 + \sin^2 x) \right]^{\frac{1}{2(e^{x^2} - 1 + \sin^2 x)}} \right\}^{\frac{2(e^{x^2} - 1 + \sin^2 x)}{x^2}}$$

$$= (e)^{2 \lim_{x \rightarrow 0} \left(\frac{e^{x^2} - 1}{x^2} + \frac{\sin^2 x}{x^2} \right)} = e^4$$

ដូចនេះ $F = \lim_{x \rightarrow 0} (2e^{x^2} - \cos 2x)^{\frac{1}{x^2}} = e^4 \text{ ។}$

$G = \lim_{x \rightarrow 0} \left(\frac{e^x + e^{3x}}{2} \right)^{\frac{1}{x}}$ មានរាងមិនកំណត់ 1^∞

$$= \lim_{x \rightarrow 0} \left(1 + \frac{e^x + e^{3x} - 2}{2} \right)^{\frac{1}{x}}$$

$$= \lim_{x \rightarrow 0} \left\{ \left[1 + \left(\frac{e^x + e^{3x} - 2}{2} \right) \right]^{\frac{2}{e^x + e^{3x} - 2}} \right\}^{\frac{e^x + e^{3x} - 2}{2x}}$$

$$= (e)^{\frac{1}{2} \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} + 3 \frac{e^{3x} - 1}{3x} \right)} = e^2$$

ដូចនេះ $G = \lim_{x \rightarrow 0} \left(\frac{e^x + e^{3x}}{2} \right)^{\frac{1}{x}} = e^2 \text{ ។}$

$$H = \lim_{x \rightarrow 0} \left(\frac{2 \cos 2x}{1 + \cos 2x} \right)^{\frac{1}{x^2}} \text{ មានរាងមិនកំណត់ } 1^\infty$$

$$= \lim_{x \rightarrow 0} \left(1 - \frac{1 - \cos 2x}{1 + \cos 2x} \right)^{\frac{1}{x^2}} = \lim_{x \rightarrow 0} \left(1 - \frac{2 \sin^2 x}{2 \cos^2 x} \right)^{\frac{1}{x^2}}$$

$$= \left[\left(1 - \tan^2 x \right)^{\frac{-1}{\tan^2 x}} \right]^{\frac{-\tan^2 x}{x^2}} = e^{-1} = \frac{1}{e}$$

ដូចនេះ $H = \lim_{x \rightarrow 0} \left(\frac{2 \cos 2x}{1 + \cos 2x} \right)^{\frac{1}{x^2}} = \frac{1}{e}$ ។

$$I = \lim_{x \rightarrow 0} \left(\frac{1 + x \sin 2x}{\cos 2x} \right)^{\frac{1}{x^2}} \text{ មានរាងមិនកំណត់ } 1^\infty$$

$$= \lim_{x \rightarrow 0} \left(1 + \frac{1 - \cos 2x + x \sin 2x}{\cos 2x} \right)^{\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0} \left(1 + \frac{2 \sin^2 x + x \sin 2x}{\cos x} \right)^{\frac{1}{x^2}}$$

$$= \left(e \right)^{\lim_{x \rightarrow 0} \frac{2 \sin^2 x + x \sin 2x}{x^2 \cos x}}$$

$$= \left(e \right)^{\lim_{x \rightarrow 0} \frac{2 \left(\frac{\sin x}{x} \right)^2 + \frac{\sin 2x}{2x} \times 2}{\cos x}} = e^4 \text{ ។}$$

ដូចនេះ $I = \lim_{x \rightarrow 0} \left(\frac{1 + x \sin 2x}{\cos 2x} \right)^{\frac{1}{x^2}} = e^4$ ។

$$J = \lim_{x \rightarrow 0} \left(\frac{\sqrt{1 + x^2}}{\cos x} \right)^{\frac{1}{x^2}} \text{ មានរាងមិនកំណត់ } 1^\infty$$

$$= \lim_{x \rightarrow 0} \left(1 + \frac{\sqrt{1+x^2} - \cos x}{\cos x} \right)^{\frac{1}{x^2}} \quad \text{។}$$

$$= \left(e \right)^{\lim_{x \rightarrow 0} \frac{\sqrt{1+x^2} - \cos x}{x^2 \cos x}} = \left(e \right)^{\lim_{x \rightarrow 0} \frac{1 + \frac{\sin^2 x}{x^2}}{\cos x (\sqrt{1+x^2} + \cos x)}} = e$$

ដូចនេះ $J = \lim_{x \rightarrow 0} \left(\frac{\sqrt{1+x^2}}{\cos x} \right)^{\frac{1}{x^2}} = e \quad \text{។}$

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